

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

November 2011 Examinations

INDICATIVE SOLUTIONS

Q1 The probability distribution is

$$\begin{array}{rcccl} X : & 0 & 1 & 2 & \\ P_x : & p & 1-2p & p & \end{array}$$

$$E(X) = \sum_{x=0}^2 x P_x = 0 * p + 1 * (1-2p) + 2 * p = 1$$

$$E(X^2) = \sum_{x=0}^2 x^2 P_x = 0^2 * p + 1^2 * (1-2p) + 2^2 * p = 1 + 2p$$

$$\begin{aligned} \therefore \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= (1+2p) - 1 \\ &= 2p \end{aligned}$$

Since $0 \leq p \leq 0.5$, $\text{Var}(X)$ is maximum when $p = 0.5$

[3]

Q2 We are given $f(x) = \begin{cases} \frac{1}{18}(3+2x) & \text{for } 2 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$

We note:

- $f(x) \geq 0$ for all values of x
- $\int_{-\infty}^{\infty} f(x) dx = \int_2^4 \frac{1}{18}(3+2x) dx$

$$= \frac{1}{18} [3(4-2) + \frac{2}{2}(4^2 - 2^2)]$$

$$= 1$$

Hence, $f(x)$ is a probability density function

$$\begin{aligned} \text{Required probability} &= P[2 \leq X \leq 3] \\ &= \int_2^3 \frac{1}{18}(3+2x) dx \\ &= \frac{1}{18} [3(3-2) + \frac{2}{2}(3^2 - 2^2)] \\ &= \frac{4}{9} \end{aligned}$$

[4]

Q3 $f(x,y) = \frac{e^{-yx^2/2}}{\sqrt{2\pi/y}} \cdot ye^{-y}, \quad -\infty < x < \infty, y > 0$

(a) By inspection, it is not hard to see that $f(x,y) = f_Y(y) * f_{X|Y}(x|y)$ where:

$$f_Y(y) = ye^{-y}, y \geq 0 \quad \& \quad f_{X|Y}(x|y) = \frac{e^{-yx^2/2}}{\sqrt{2\pi/y}}, -\infty < x < \infty$$

Equivalently,

$$f_Y(y) = \frac{1^2}{\Gamma(2)} \cdot y^{2-1} e^{-1 \cdot y}, y \geq 0$$

$$\& \quad f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi}(y^{-0.5})} \cdot e^{-\frac{1}{2} \left(\frac{x-0}{y^{-0.5}} \right)^2}, -\infty < x < \infty$$

Thus, using the actuarial tables we can state

$$Y \sim \text{Gamma}(\alpha = 2, \lambda = 1)$$

$$\& \quad X | Y = y \sim \text{Normal}(\mu = 0, \sigma^2 = y^{-1})$$

[Full credit available only if the names and parameters of the distributions are identified]

(b) As $Y \sim \text{Gamma}(\alpha = 2, \lambda = 1)$

$$E(Y) = \frac{\alpha}{\lambda} = 2$$

$$\text{Var}(Y) = \frac{\alpha}{\lambda^2} = 2$$

(c) As the conditional distribution $X|Y = y \sim N\left(0, \frac{1}{y}\right)$

$$E(X|Y = y) = 0$$

$$\text{Var}(X|Y = y) = \frac{1}{y}$$

(d) $E(X) = E[E(X|Y)]$

$$= E(0)$$

$$= 0$$

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$$

$$= E(1/Y) + \text{Var}(0)$$

$$= \int_0^{\infty} \frac{1}{y} \cdot ye^{-y} dy + 0 = \int_0^{\infty} e^{-y} dy = 1.$$

[10]

The problem required students to identify type of distributions by looking at the joint distribution and derive the moments using the formula given in the actuarial tables. But, full credit is also available for students who solve this problem correctly from first principles using integration.

Q4 Let N be the number of clubs accepted

X be the number of members of a selected club

S be the total persons appearing.

From the information given in the question, it is clear that $N \sim \text{Binomial} (n=1,000, p=0.20)$

Thus,

- $E(N) = (1,000) (0.20) = 200$
 - $\text{Var}(N) = (1,000) (0.20) (0.80) = 160$
- Further, $E(X) = 20, \text{Var}(X) = 20$

$$\begin{aligned}\text{Therefore, } E(S) &= E(N) E(X) \\ &= (200) (20) \\ &= 4,000\end{aligned}$$

$$\begin{aligned}\text{Var}(S) &= E(N) \text{Var}(X) + \text{Var}(N) [E(X)]^2 \\ &= (200) (20) + (160) (20)^2 \\ &= 68,000\end{aligned}$$

$$\begin{aligned}\text{Hence, the annual budget for persons appearing on the show will be} \\ &= (10).E(S) + (10).\sqrt{\text{Var}(S)} \\ &= (10)(4,000) + (10)\sqrt{68,000} \\ &\approx 42,608\end{aligned}$$

[5]

Q5 (a) Assuming all the data values in each interval are equal to the mid-point, we get observations of 5.08, 5.09,, 5.15.

Intervals	x_i	f_i	$x_i f_i$	$x_i^2 f_i$
5.075 - 5.085	5.08	1	5.08	25.8064
5.085 - 5.095	5.09	3	15.27	77.7243
5.095 - 5.105	5.10	3	15.30	78.0300
5.105 - 5.115	5.11	7	35.77	182.7847
5.115 - 5.125	5.12	10	51.20	262.1440
5.125 - 5.135	5.13	23	117.99	605.2887
5.135 - 5.145	5.14	56	287.84	1,479.4976
5.145 - 5.155	5.15	97	499.55	2,572.6825
		200	1,028.00	5,283.9582

$$\begin{aligned}\text{Mean of the sample} &= \mu_R = \frac{\sum x_i f_i}{\sum f_i} \\ &= \frac{1028}{200} \\ &= 5.14\end{aligned}$$

Variance of the sample

$$\begin{aligned}
 s_R^2 &= \frac{\sum x_i^2 f_i - (\sum f_i) \mu_R^2}{\sum f_i - 1} \\
 &= \frac{1}{199} [5283.9582 - (200)(5.14)^2] \\
 &= \frac{0.0382}{199} \\
 &= 0.00019196
 \end{aligned}$$

Hence, the standard deviation of the sample

$$\begin{aligned}
 s_R &= \sqrt{0.00019196} \\
 &= 0.013855
 \end{aligned}$$

(b)[i] Let W denote the weight of the package.

Now a package is rejected as underweight if $W < 5.155$

To estimate the mean μ for the whole distribution whose (estimated) variance is

$$\begin{aligned}
 \sigma^2 &= (4s_R)^2 = (16)(0.00019196) \\
 &= 0.00307
 \end{aligned}$$

We know that

$$\begin{aligned}
 0.05 &= P[W < 5.155] \\
 &= P\left[\frac{W - \mu}{\sigma} < \frac{5.155 - \mu}{\sigma}\right]
 \end{aligned}$$

Using the value of the 5% point of $N(0,1)$ we need

$$\begin{aligned}
 \frac{5.155 - \mu}{\sigma} &= -1.645 \\
 \mu &= 5.155 + 1.645\sigma \\
 &= 5.155 + 1.645 \cdot \sqrt{0.00307} \\
 &= 5.246
 \end{aligned}$$

(ii) Let μ_A be the mean weight of the packages not rejected.

Then,

$$\begin{aligned}
 \mu &= 0.05\mu_R + 0.95\mu_A \\
 \mu_A &= \frac{\mu - 0.05\mu_R}{0.95} \\
 &= \frac{5.246 - (0.05)(5.14)}{0.95} = 5.252
 \end{aligned}$$

[10]

Q6 Let n be the (unknown) number of light bulbs to be purchased.

Let $X_1, X_2, \dots, X_n$ be their respective lifetimes. Denote $S_n = \sum_{i=1}^n X_i$ as the total lifetime of all n bulbs.

We need to choose 'n' minimal so that

$$P[S_n \geq 40] \geq 0.9772$$

We are given: $\left. \begin{array}{l} E(X_i) = 3 \\ \text{Var}(X_i) = 1 \end{array} \right\} \text{ for } i = 1, 2 \dots n$

Thus: $\left. \begin{array}{l} E(S_n) = 3n \\ \text{Var}(S_n) = n \end{array} \right\}$

Now, using the Central Limit Theorem:

$$P[S_n \geq 40] = P\left[\frac{S_n - 3n}{\sqrt{n}} \geq \frac{40 - 3n}{\sqrt{n}}\right] \approx 1 - \Phi\left(\frac{40 - 3n}{\sqrt{n}}\right)$$

Note: No continuity correction is required as S_n takes values over $[0, \infty)$

This is equal to 0.9772 when $\frac{40 - 3n}{\sqrt{n}} = -2$

$$\Rightarrow 3n - 2\sqrt{n} - 40 = 0$$

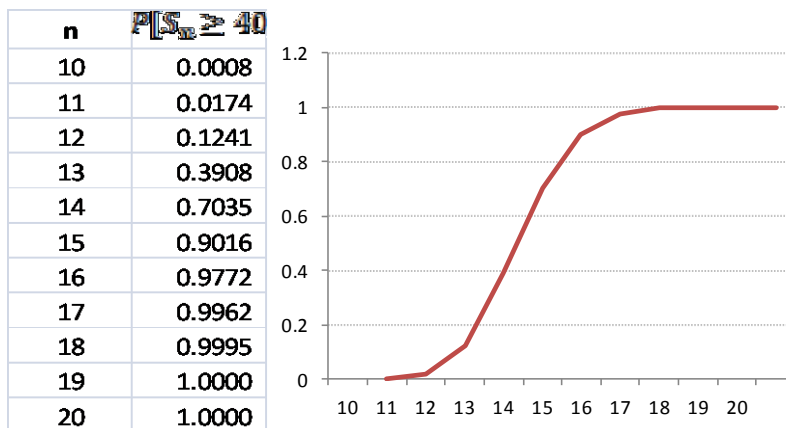
Setting $x = \sqrt{n}$, the latter equation becomes: $3x^2 - 2x - 40 = 0$.

Solving (ignoring the negative solution) gives: $x = \frac{1}{6}(2 + \sqrt{4 + 480}) = 4$

$\Rightarrow n = x^2 = 16$ is the number sought.

[5]

Note, $n = 16$ is the minimum value of 'n' for which the inequality $P[S_n \geq 40] \geq 0.9772$ is satisfied:



Q7 We have $S = X_1 + X_2 \dots + X_{10}$ as the sum of the 10 numbers obtained.

We first compute the moment generating function of an individual X_i

$$\begin{aligned}
 M_X(t) &= E(e^{tX}) \\
 &= \sum_{k=1}^4 e^{tk} P(X=k) \\
 &= \frac{1}{4} e^t \sum_{k=0}^3 e^{tk} & [\because P(X=k) = \frac{1}{4} \quad \forall k] \\
 &= \frac{e^t(1-e^{4t})}{4(1-e^t)}
 \end{aligned}$$

[using the finite geometric series formula $\sum_{k=0}^n r^k = \frac{1-r^{n+1}}{1-r}$]

Alternately this can be derived using the formula given in page 10 of the actuarial tables:

This is a discrete uniform random variable with parameters:

- $a = 1$
- $b = 4$
- $h = 1$

Thus,
$$p(x) = \frac{1}{4-1+1}$$

$$= \frac{1}{4} \quad \text{for } x = 1, 2, 3, 4$$

Hence:

$$\begin{aligned}
 \text{MGF: } M_X(t) &= \frac{1}{4-1+1} \cdot \left[\frac{e^{(4+1)t} - e^t}{e^t - 1} \right] \\
 &= \frac{e^t(1 - e^{4t})}{4(1 - e^t)}
 \end{aligned}$$

Now, the moment generating function of a sum of independent random variables is the product of the individual moment generating functions.

Thus, we get

$$M_s(t) = \prod_1^{10} M_{X_i}(t) = \left[\frac{e^t - e^{5t}}{4(1 - e^t)} \right]^{10}$$

[3]

Q8 Estimator S of θ is defined as

$$S = \begin{cases} \frac{1}{2}(X+Y) & \text{if } X \neq Y \\ X - \frac{1}{3} & \text{if } X = Y \end{cases}$$

(a) We know X and Y has the following probability mass function:

Value:	$\theta-1$	$\theta+1$
Probability:	$\frac{1}{3}$	$\frac{2}{3}$

Thus, the joint probability mass function (using the fact that they are independent) is

		Y	
		$\theta - 1$	$\theta + 1$
X	$\theta - 1$	1/9	2/9
	$\theta + 1$	2/9	4/9

$$\text{If } X \neq Y, \text{ then } X + Y = (\theta - 1) + (\theta + 1) \text{ or } (\theta + 1) + (\theta - 1) \\ = 2\theta$$

$$\Rightarrow S = \frac{1}{2}(X + Y) = \theta$$

$$\text{Thus, if } X \neq Y, P(S = \theta) = P(X = \theta - 1, Y = \theta + 1) + P(X = \theta + 1, Y = \theta - 1) \\ = \frac{2}{9} + \frac{2}{9} \\ = \frac{4}{9}$$

$$\text{If } X = Y, \text{ then } S = X - \frac{1}{3} = \begin{cases} \theta - \frac{4}{3} & \text{when } X = \theta - 1, Y = \theta - 1 \\ \theta + \frac{2}{3} & \text{when } X = \theta + 1, Y = \theta + 1 \end{cases}$$

Thus,

$$P\left(S = \theta - \frac{4}{3}\right) = P(X = \theta - 1, Y = \theta - 1) = \frac{1}{9}$$

$$P\left(S = \theta + \frac{2}{3}\right) = P(X = \theta + 1, Y = \theta + 1) = \frac{4}{9}$$

(b) We have the random variable S taking the values

$$S = \begin{cases} \theta & \text{with probability } 4/9 \\ \theta - \frac{4}{3} & \text{with probability } 1/9 \\ \theta + \frac{2}{3} & \text{with probability } 4/9 \end{cases}$$

Or, equivalently the random variable $S - \theta$ takes the values

$$S - \theta = \begin{cases} 0 & \text{with probability } 4/9 \\ -\frac{4}{3} & \text{with probability } 1/9 \\ +\frac{2}{3} & \text{with probability } 4/9 \end{cases}$$

Now, Bias of S as an estimator of θ :

$$\begin{aligned}
 E(S) - \theta &= E(S - \theta) \\
 &= 0 \cdot \frac{4}{9} + \left(-\frac{4}{3}\right) \cdot \frac{1}{9} + \frac{2}{3} \cdot \frac{4}{9} \\
 &= -\frac{4}{27} + \frac{8}{27} \\
 &= \frac{4}{27} \neq 0 \quad \Rightarrow S \text{ is a **biased** estimator of } \theta
 \end{aligned}$$

(c) The mean square error (MSE) of S as an estimator of θ

$$\begin{aligned}
 &= E[(S - \theta)^2] \\
 &= 0^2 \cdot \frac{4}{9} + \left(-\frac{4}{3}\right)^2 \cdot \frac{1}{9} + \left(\frac{2}{3}\right)^2 \cdot \frac{4}{9} \\
 &= \frac{16}{81} + \frac{16}{81} \\
 &= \frac{32}{81}
 \end{aligned}$$

Alternately, $E(S)$ and $MSE(S)$ can be derived by computing the moments of S directly

(d) Comparing the two competing estimators of θ

	S	T	
Bias	$\frac{4}{27}$	0	[Bias(T)=0 as T is unbiased]
MSE	$\frac{32}{81}$	$\frac{36}{81}$	[Var(T)=4/9 & as it is unbiased, MSE(T) = Var(T)]

Since, S has a lower mean square error than T Shriya should use the estimator S for guessing the value of θ in spite of it being a biased estimator unlike T.

[10]

Q9. (a) An unbiased estimator of σ_x^2

$$\begin{aligned}
 &= \frac{1}{11-1} \sum_{i=1}^{11} (x_i - \bar{x})^2 \\
 &= \frac{1}{10} (12.3251) \\
 &= 1.23251
 \end{aligned}$$

An unbiased estimator of σ_y^2

$$\begin{aligned}
 &= \frac{1}{11-1} \sum_{i=1}^{11} (y_i - \bar{y})^2 \\
 &= \frac{1}{10} (1.7101) \\
 &= 0.17101
 \end{aligned}$$

(b) The pivotal quantity is

$$\frac{s_x^2/s_y^2}{\sigma_x^2/\sigma_y^2} \sim F_{11-1, 11-1} = F_{10, 10}$$

Hence, a 95% equal-tailed confidence interval is given by:

$$\frac{s_x^2/s_y^2}{F_{10, 10; 0.025}} < \frac{\sigma_x^2}{\sigma_y^2} < \frac{s_x^2/s_y^2}{F_{10, 10; 0.975}}$$

Now,

$$\frac{s_x^2}{s_y^2} = \frac{\sum_{i=1}^{11} (x_i - \bar{x})^2}{\sum_{i=1}^{11} (y_i - \bar{y})^2} = \frac{12.3251}{1.7101} = 7.20724$$

$$\left. \begin{array}{l} F_{10, 10; 0.975} = 0.269 \\ F_{10, 10; 0.025} = 3.717 \end{array} \right\} \text{ as per the values given in the question}$$

Thus, the confidence interval is

$$\frac{7.20724}{3.717} < \frac{\sigma_x^2}{\sigma_y^2} < \frac{7.20724}{0.269} \Rightarrow (1.9390, 26.7927)$$

(c) We want to test

$$H_0: \frac{\sigma_x^2}{\sigma_y^2} = c \text{ against } H_1: \frac{\sigma_x^2}{\sigma_y^2} \neq c \text{ for some constant 'c' at the 5\% level.}$$

We have obtained a 95% equal-tailed confidence interval of σ_x^2/σ_y^2 as (1.9390, 26.7927). If we find 'c' lying within the above confidence interval, we can be confident at 5% level that H_0 can't be rejected. Otherwise we will accept H_1 .

If $c = 30$, we see it does not lie within the confidence interval. So at 5% level, we cannot accept H_0 ;

$$\sigma_x^2/\sigma_y^2 = 30$$

[10]

Q10.(a) The likelihood function is given by

$$\begin{aligned} L(\theta) &= [(1 - \theta)^2]^{34} \cdot [2\theta(1 - \theta)^2]^{25} \cdot [\theta^2(3 - 2\theta)]^{275} \\ &\propto \theta^{25+2(275)} \cdot (1 - \theta)^{2(34+25)} \cdot (3 - 2\theta)^{275} \\ &\propto \theta^{575} \cdot (1 - \theta)^{118} \cdot (3 - 2\theta)^{275} \end{aligned}$$

The log likelihood function is given by

$$l(\theta) = \ln[L(\theta)]$$

$$\propto 575 \ln \theta + 118 \ln(1 - \theta) + 275 \ln(3 - 2\theta)$$

Differentiating,

$$\begin{aligned}\frac{\partial l(\theta)}{\partial \theta} &= \frac{575}{\theta} - \frac{118}{1-\theta} - \frac{550}{3-2\theta} \\ \frac{\partial l(\theta)}{\partial \theta} = 0 &\Rightarrow \frac{118}{1-\theta} + \frac{550}{3-2\theta} = \frac{575}{\theta} \\ &\Rightarrow 1936\theta^2 - 3779\theta + 1725 = 0\end{aligned}$$

Solving for θ :

$$\begin{aligned}\theta &= \frac{3779 \pm \sqrt{14280841 - 13358400}}{3872} \\ &= \frac{3779 \pm 960.43792}{3872} \\ &= 1.224, 0.728\end{aligned}$$

Second derivative $\frac{\partial^2 l(\theta)}{\partial \theta^2} = -\frac{575}{\theta^2} - \frac{118}{(1-\theta)^2} - \frac{1100}{(3-2\theta)^2} < 0$ which confirms maximum

Since $0 < \theta < 1$, we have $\hat{\theta} = 0.728$

Hence, the maximum likelihood estimate of θ is 0.728

Full credit is also available for students who solve this problem correctly using alternate approaches like showing 0.728 is a root of the quadratic equation as long as they establish why 0.728 is the MLE.

(b) We are testing the following hypothesis using a χ^2 goodness of fit test:

H_0 : model provides a good fit to the data against

H_1 : model does not provide a good fit to the data.

Using the maximum likelihood estimate of θ , we first estimate the probabilities of total number infected:

	Total number infected		Probability	
	1	$(1 - \theta)^2 = 0.073984$		
	2	$2\theta(1 - \theta)^2 = 0.107721$		
	3	$\theta^2(3 - 2\theta) = 0.818295$		

# Infected	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2 / E_i$
1	34	24.71	9.29	3.493
2	25	35.98	-10.98	3.351
3	275	273.31	1.69	0.010
	334	334		6.854

Here $E_i = 334 * \text{Prob}(\text{Total infected} = i)$ $i=1,2,3$ gives the expected frequencies

The test statistic is,

$$\chi^2 = \sum_{i=1}^3 \frac{(O_i - E_i)^2}{E_i} = 6.854$$

This follows a chi-square distribution with degrees of freedom = $3 - 1 - 1 = 1$

Since the observed value of the test statistics is more than the 5% critical value of 3.841, we have insufficient evidence at the 5% level to accept H_0 .

We therefore conclude that the model does **not** provide a good fit to these data.

[12]

Q11 (a) We have

$$\begin{aligned} SS_T &= \sum_{i=1}^6 \sum_{j=1}^5 y_{ij}^2 - \frac{1}{30} \left(\sum_{i=1}^6 \sum_{j=1}^5 y_{ij} \right)^2 \\ &= 661796 - \frac{1}{30} (4426)^2 \\ &= 8813.47 \end{aligned}$$

$$\begin{aligned} SS_B &= \frac{1}{5} \left[\sum_{i=1}^6 \left(\sum_{j=1}^5 y_{ij} \right)^2 \right] - \frac{1}{30} \left(\sum_{i=1}^6 \sum_{j=1}^5 y_{ij} \right)^2 \\ &= \frac{1}{5} [748^2 + 657^2 + 826^2 + 741^2 + 710^2 + 744^2] - \frac{1}{30} (4426)^2 \\ &= 3046.67 \end{aligned}$$

$$\begin{aligned} SS_R &= SS_T - SS_B = 8813.47 - 3046.67 \\ &= 5766.80 \end{aligned}$$

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Treatments	5	3046.67	609.3	2.54
Residual	24	5766.8	240.3	
	29	8813.47		

From tables $F_{5,24}(5\%) = 2.621$

As observed $F < 2.621$, we have sufficient evidence to state that there are **no** significant differences at the 5% level between mean premiums being charged by each company.

(b)i] Estimate of variance from the ANOVA

$$\hat{\sigma}^2 = 240.3$$

For comparing company B and C, the t-test is given by:

$$H_0: \mu_B = \mu_C \text{ against } H_1: \mu_B \neq \mu_C$$

The t-statistic is given by

$$t = \frac{|\bar{y}_B - \bar{y}_C|}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{5} + \frac{1}{5} \right)}} \text{ which follows a t-distribution with 24 degrees of freedom}$$

$$\begin{aligned} \text{Observed } t &= \frac{|131.4 - 165.2|}{\sqrt{240.30 \cdot \frac{2}{5}}} \\ &= \frac{33.8}{9.8} \\ &= 3.45 \end{aligned}$$

From tables, $t_{24}(0.5\%) = 2.797$

As observed $t > 2.797$, we have sufficient evidence to state that there is significant difference at the 1% level (two-sided).

(ii) There is no contradiction.

It is wrong to pick out the largest and the smallest of a set of treatment means, test for significance, and then draw conclusions about the set.

Even if $H_0: \mu_i \text{ s all equal}$ is true, the largest and smallest sample means would, of course, differ.

[10]

Q12. (a) The linear regression model is given by

$$y_i = \alpha + \beta(x_i - \bar{x}) + e_i, \quad i=1, 2, \dots, 12$$

with e_i are independent $N(0, \sigma^2)$ error variables

Equivalently,

$$\begin{aligned} y_i &= (\alpha - \beta\bar{x}) + \beta x_i + e_i \\ &= a + b x_i + e_i, \quad i=1, 2, \dots, 12 \end{aligned}$$

where $a = \alpha - \beta\bar{x}$, $b = \beta$

i. Using the results from the actuarial tables, the least square estimates for a and b will be given by:

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

$$\hat{b} = \frac{S_{xy}}{S_{xx}} \quad \text{where} \quad \left. \begin{aligned} S_{xy} &= \sum_{i=1}^{12} x_i y_i - 12\bar{x}\bar{y} \\ S_{xx} &= \sum_{i=1}^{12} x_i^2 - 12\bar{x}^2 \end{aligned} \right\}$$

Thus the least square estimates of α and β will be given by:

$$\begin{aligned}\hat{\beta} &= \hat{b} = \frac{S_{xy}}{S_{xx}} \\ \hat{\alpha} &= \hat{a} + \hat{\beta}\bar{x} \\ &= (\bar{y} - \hat{b}\bar{x}) + \hat{\beta}\bar{x} \\ &= \bar{y}\end{aligned}$$

ii. Estimate of $\sigma^2 = \hat{\sigma}^2$

$$= \frac{1}{n-2} (S_{yy} - \frac{S_{xy}^2}{S_{xx}}) \quad \text{where } S_{yy} = \sum_{i=1}^{12} y_i^2 - 12\bar{y}^2$$

The problem required students to transform the given regression model into one in the actuarial tables and thereafter use the results given to derive the least square estimates of the parameters. But, full credit is also available for students who solve this problem correctly from first principles using minimizing least squares principles.

(b) The regression line is given by

$$\hat{y} = \hat{\alpha} + \hat{\beta}(x - \bar{x})$$

$$\text{Here: } \hat{\alpha} = \bar{y} = \frac{1}{12} (716) = 59.67$$

$$\begin{aligned}\hat{\beta} &= \frac{S_{xy}}{S_{xx}} \\ &= \frac{27503 - 12\left(\frac{716}{12}\right)\left(\frac{453}{12}\right)}{17372 - 12\left(\frac{453}{12}\right)^2} \\ &= \frac{474.00}{271.25} \\ &= 1.75\end{aligned}$$

Therefore the regression line:

$$\hat{y} = 59.67 + 1.75\left(x - \frac{453}{12}\right)$$

$$\text{or } \hat{y} = -6.39 + 1.75x$$

(c) We want to test for

$$H_0: \beta = 0 \text{ against } H_1: \beta > 0$$

The t-statistic to test this is given by

$$\begin{aligned}t &= \frac{\beta - 0}{\sqrt{\hat{\sigma}^2 / S_{xx}}} \sim t_{12-2} \\ &= \frac{\hat{\beta}}{\hat{\sigma} / \sqrt{S_{xx}}} \sim t_{10} \text{ distribution.}\end{aligned}$$

We have $\hat{\beta} = 1.75$

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{10} \left[\left\{ 43827 - 12 \left(\frac{716}{12} \right)^2 \right\} - \frac{(474)^2}{271.25} \right] \\ &= \frac{1}{10} \left[1105.67 - \frac{(474)^2}{271.25} \right] \\ &= 27.74\end{aligned}$$

$$\begin{aligned}\text{Observed value } t &= \frac{1.75}{\sqrt{27.74/271.25}} \\ &= 5.465\end{aligned}$$

From tables, $t_{10}(5\%) = 1.812$

As observed $t > 1.812$, we have sufficient evidence to reject H_0 at 5% level of significance.

(d) We are told that y_i s were wrongly recorded. Instead the recorded values should be $10y_i$.

Denote $y_i^* = 10y_i$, $i=1, 2, \dots, 12$

i. We will have

$$\begin{aligned}S_{xy}^* &= \sum_1^{12} x_i y_i^* - 12 \bar{x} \bar{y}^* \\ &= 10 \left(\sum_1^{12} x_i y_i - 12 \bar{x} \bar{y} \right) \quad [\because \bar{y}^* = 10 \bar{y}] \\ &= 10 S_{xy}\end{aligned}$$

$$\begin{aligned}\text{Therefore, } \hat{\beta}^* &= \frac{S_{xy}^*}{S_{xx}} \\ &= 10 \frac{S_{xy}}{S_{xx}} = 10 \hat{\beta}\end{aligned}$$

ii. We will have

$$\begin{aligned}S_{yy}^* &= \sum_1^{12} y_i^{*2} - 12 \bar{y}^{*2} \\ &= 100 \left(\sum_1^{12} y_i^2 - 12 \bar{y}^2 \right) \quad [\because \bar{y}^* = 10 \bar{y}] \\ &= 100 S_{yy}\end{aligned}$$

$$\begin{aligned}\text{Thus, } \hat{\sigma}^{*2} &= \frac{1}{10} \left[S_{yy}^* - \frac{S_{xy}^{*2}}{S_{xx}} \right] \\ &= 100 \cdot \frac{1}{10} \left(S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right) \\ &= 100 \hat{\sigma}^2\end{aligned}$$

iii. Therefore, the revised t-statistic is given by

$$\begin{aligned}
 t^* &= \frac{\hat{\beta}^*}{\sqrt{\hat{\sigma}^{*2}/S_{xx}}} \\
 &= \frac{10 \hat{\beta}}{\sqrt{100 \hat{\sigma}^2/S_{xx}}} \\
 &= \frac{\hat{\beta}}{\sqrt{\hat{\sigma}^2/S_{xx}}} \\
 &= t \quad \text{as earlier}
 \end{aligned}$$

Therefore, the t-test for the given problem in part (c) will **not** change as a result.
 The conclusion will remain same, i.e., Reject $H_0: \beta = 0$ at 5% level of significance.

[18]

Full credit is also available for students who solve part (d) correctly from first principles using minimizing least squares principles.

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